TEXORA · LECTURE SHEET

The Draw Frame

The machine that makes sliver even: several carded slivers are fed together, drafted back to the weight of one, and in that doubling and drafting their irregularities average out, their fibres are parallelised, and their hooks are straightened.

TE-2101 · SPINNING TECHNOLOGY-I · TOPIC 4 · LEVEL 2 TERM 1

1. Objects of the draw frame

Card sliver is the most irregular product in the spinning line: its weight varies along its length, its fibres are wavy and hooked, and it still holds dust from carding. Drafting alone cannot correct weight variation — it copies feed irregularity into the delivery. The draw frame therefore combines drafting with **doubling**: several slivers are fed together and drafted down to about the weight of one. Its objects are:

- **Drafting (attenuation)** — to reduce the combined feed to the required delivered sliver weight by roller drafting, in a controlled way that moves fibres evenly.
- **Doubling for evenness** — to lay several slivers together so that thick places in some coincide with thin places in others and average out; doubling is the draw frame's main evenness tool (Section 5).
- **Blending** — slivers from different cards (and, in a blend line, of different fibres) are mixed intimately by the doubling, so composition variations are spread uniformly through the delivery.
- **Parallelisation and hook removal** — the draft straightens the wavy, hooked fibres left by the card and lays them parallel to the sliver axis; the direction of hooks decides how many draw-frame passages are needed and in which order machines follow (Section 4).
- **Dust removal** — suction at the drafting zone removes dust and fly liberated as the slivers are opened by drafting (a secondary but useful cleaning).
- **Regular sliver delivery** — to deliver a sliver of controlled weight, coiled in cans, fit to feed the next process (a second draw-frame passage, the comber line, or the speed frame).

Passages are described as **breaker** (first) and **finisher** (second) drawing. Carded routes normally use two passages; combed routes use more (drawing before combing to prepare laps, and one or two after combing). Each passage improves evenness but adds cost and a fresh chance of machine faults, so the number of passages is an economic as well as a technical choice.

2. Passage of material through the draw frame

1. **Cans at the creel.** Six or eight cans of card (or previous-passage) sliver stand behind the machine; each sliver is withdrawn from the top of its coil without disturbing the sliver below.
2. **Guide rollers and feed guides.** Each sliver passes over a positively driven guide roller and through its own guide eye, keeping the slivers separate, side by side, under light equal tension until they enter the drafting system together as a flat fibre sheet.
3. **Drafting system.** The combined slivers pass through successive roller pairs — back, middle and front in a 3-over-3 system — each pair faster than the one before, attenuating the sheet to the target weight.
4. **Web.** From the front roller nip the fibres emerge as a wide, thin, transparent web — the material at its weakest and most delicate.
5. **Trumpet (condenser).** The web is gathered through a trumpet from its full width into a round sliver, restoring cohesion.
6. **Calendar rollers.** A pair of smooth rollers compresses the sliver and expels air, giving a firm, measurable delivery.
7. **Coiler and can.** The coiler head lays the sliver in regular overlapping coils in a rotating can, ready for the next machine. Modern frames have two deliveries side by side, each with its own creel, drafting system and coiler.

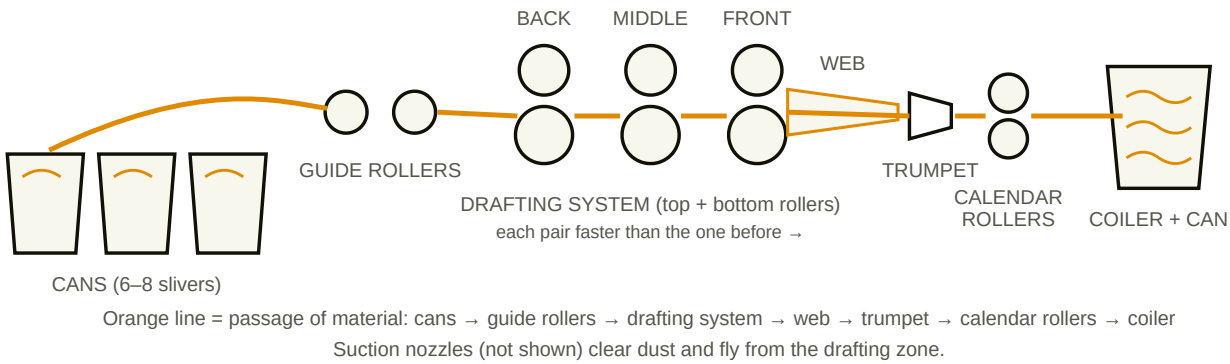


Figure 1 — Passage of material through a draw frame, side view (not to scale). Six to eight slivers are fed together and drafted back to approximately the weight of one.

3. The drafting system

3.1 How roller drafting works

Each drafting unit is a pair of rollers forming a **nip** that grips the fibre sheet: bottom rollers are fluted steel, positively driven; top rollers carry synthetic rubber cots and are pressed down onto them. If the front pair's surface speed is higher than the back pair's, a fibre seized by the front nip is pulled forward faster than the material behind it, and the sheet between the nips is stretched thinner — drafted. The ratio of surface speeds is that zone's draft.

Draft of a zone = (surface speed of the front pair) ÷ (surface speed of the back pair)

A fibre between two nips is in one of three states: gripped by the back nip (moving at back speed), gripped by the front nip (moving at front speed), or — in between — **floating**: held by neither nip, carried only by friction with its neighbours. Floating fibres are the central problem of drafting: if many fibres float uncontrolled over a long distance they move in clumps, and the delivered sliver shows the periodic fault called drafting wave (Section 7). Good drafting design minimises the floating zone and controls floating fibres.

3.2 Drafting arrangements

SYSTEM	ARRANGEMENT	NOTES
3-over-3	Three top rollers over three bottom rollers (two drafting zones)	The standard modern system; simple, easy to set, good fibre control; used in this sheet's worked examples
4-over-4	Four top over four bottom rollers (three zones)	Draft shared over more, smaller zones; gentler drafting for longer or delicate fibres and higher total drafts; more rollers to maintain
5-over-3	Five top rollers over three bottom rollers	A pressure bar in the main zone presses the fibre sheet down to control floating fibres — improved main-zone control, common on high-performance frames

3.3 Roller types and weighting

Bottom rollers are fluted (longitudinally grooved) steel: the flutes give positive grip and stop the fibre sheet slipping sideways. **Top rollers (cots)** are metal cores covered with synthetic rubber of controlled hardness — soft enough to grip fibres without cutting them, hard enough to resist grooving. A cut, grooved or eccentric cot stamps a periodic thick place into every sliver it touches. Top rollers are loaded by **weighting** — dead weights on older frames, spring or pneumatic loading on modern frames — with the front nip weighted most heavily, since it holds the fibres against the full pull of drafting. Clearers and suction keep rollers free of fly that would otherwise build up into laps.

3.4 Break draft and main draft

The total draft is shared between zones, not taken in one step. The **break draft**, in the back zone, is small — typically about 1.1–1.5 — and only straightens the incoming slivers under tension, removing crimp and light hooks. The **main draft**, in the front zone, does most of the attenuation — commonly of the order of 5–8 with six to eight doublings. Total draft = break draft × main draft (two-zone, 3-over-3 system). Too much break draft wastes the front zone's fibre control; too little leaves the sheet under-tensioned and wavy entering the main zone.

3.5 Roller setting

The **roller setting** is the distance between the nip points of successive roller pairs, and it is governed by fibre length. **The rule:** each zone's setting must be slightly *longer* than the effective (staple) length of the fibres being drafted.

Roller setting ≈ staple length + a small margin (commonly about 4–10 mm, larger in the back zone)

If the setting is *shorter* than the longest fibres, a fibre gripped by both nips at once is broken or dragged violently — breakage, neps and lapping follow. If it is *too wide*, the floating zone lengthens, fibres drift in clumps, and drafting wave and irregularity increase. The back (break) zone is set wider than the front (main) zone, because the thicker, less parallel feed needs more freedom.

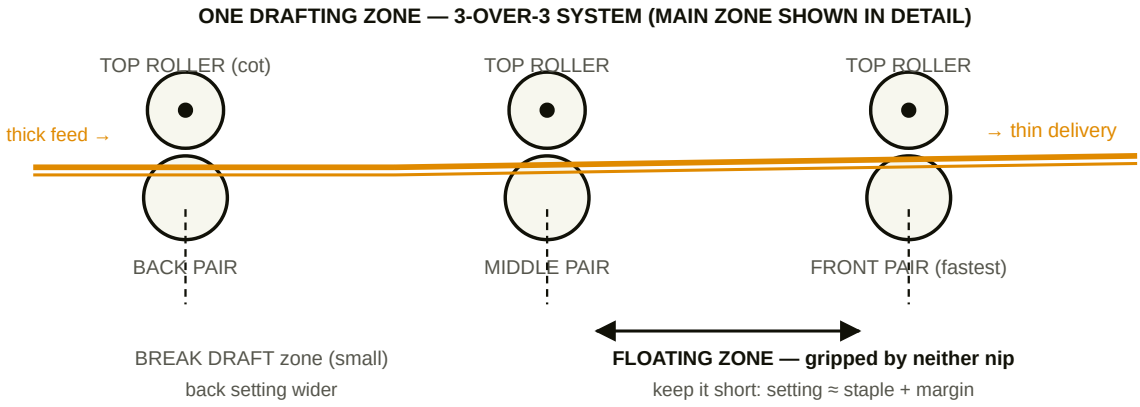


Figure 2 — Drafting zones of a 3-over-3 system. A fibre is controlled while a nip grips it and drifts by friction alone in the floating zone between nips — the zone whose length the roller setting controls.

4. Parallelisation and hooks — leading and trailing

Most fibres in card sliver carry a hook at one end, formed during transfer from cylinder to doffer. A hook is named by its direction *relative to travel*: a **leading hook** points forward, a **trailing hook** backward. Card sliver contains a majority of **trailing hooks**.

Drafting straightens a hook only when it is at the fibre's **trailing** end: the front nip grips the straight leading portion and pulls, drawing the held-back hook out straight behind it. A **leading hook** is carried

through folded — the nip grips behind it and doubles it back rather than opening it. Two consequences follow:

- Material should be presented to each drafting machine so that the **majority hook direction is trailing**. Since card sliver already has mainly trailing hooks, the first (breaker) draw frame straightens most of them.
- Every passage through a coiler and can, and every drafting, **reverses** which end of the sliver — and therefore which hook orientation — leads at the next machine. A second (finisher) passage is thus not a mere repetition: the fibres that kept leading hooks in the first passage present trailing hooks in the second, so hooks missed the first time are straightened the second time. This reversal principle also explains why an **even number of processes between card and ring frame** is the traditional rule — the roving must reach the ring frame with its majority hooks trailing so that ring-frame drafting straightens them.

5. Doubling and evenness

When n slivers are combined, a thick place in one is laid against average or thin places in the others: peaks and valleys partly cancel. If the feeds are independent, each with coefficient of variation CV, the combined strand has approximately:

$$\text{CV of combined feed} \approx \text{CV of one sliver} \div \sqrt{n}$$

with n = number of doublings. Six or eight doublings are standard at the draw frame: **6 doublings** suit lighter drafts and preserve production flexibility, **8 doublings** give the best averaging and are common on finisher passages and in combed preparation. Doubling cannot correct *long-term* variation (a whole can of heavy sliver stays heavy through the average) and it cannot correct periodic faults that are in phase in all feeds — which is why autolevelling (Section 6) and good card control remain necessary. The draft is then chosen to bring the combined feed back to the target delivery weight:

$$\text{Draw-frame draft} = (\text{number of doublings} \times \text{feed sliver weight}) \div \text{delivered sliver weight}$$

WORKED EXAMPLE — DRAW-FRAME DRAFT

Eight card slivers, each of 5 g/m, are doubled at a draw frame and the delivered sliver is 5.5 g/m. Find the total draft.

Total feed weight = 8 doublings × 5 g/m = 40 g/m.

Draft = feed weight ÷ delivery weight = 40 ÷ 5.5 = 7.2727… ≈ **7.27**.

Check by lengths: 1 m of combined feed weighs 40 g; at 5.5 g/m this mass delivers 40 ÷ 5.5 ≈ 7.27 m of sliver — the same number, as it must be.

WORKED EXAMPLE — DRAW-FRAME PRODUCTION

> One delivery of a draw frame delivers sliver at 800 m/min with a delivered sliver weight of 5 g/m. Find the production of that delivery in kg/h.
> Length delivered in one hour = 800 m/min × 60 min = 48,000 m.
> Mass delivered = 48,000 m × 5 g/m = 240,000 g = 240 kg.
> Production = 800 × 60 × 5 ÷ 1000 = **240 kg/h per delivery** (theoretical; a two-delivery machine at 90 % efficiency would deliver about 2 × 240 × 0.90 = 432 kg/h in practice).

6. Autolevelling in brief

Doubling averages short-term irregularity, but slow drifts in feed weight — a heavy can, a card going off weight — pass straight through. An **autoleveller** measures sliver thickness continuously, usually by a spring-loaded measuring roller pair at the feed, and varies the draft automatically: thick feed gets a slightly higher draft, thin feed a slightly lower one, holding delivered weight constant. **Open-loop** systems measure at the feed and correct before that length is drafted (the usual arrangement); **closed-loop** systems measure the delivery and feed the correction back. Autolevelling corrects medium- and long-term variation well; very short-term irregularity remains the job of doubling, correct settings and good rollers.

7. Draw-frame defects and their causes

DEFECT	APPEARANCE / EFFECT	PRINCIPAL CAUSES
Sliver irregularity (high CV / U%)	Delivered weight varies along the sliver; fault reappears as yarn unevenness and weak places	Too few doublings, wrong roller settings, worn or eccentric cots, uneven feed slivers, excessive draft for the fibre
Drafting wave	A regular, wavy periodicity in the sliver with a wavelength of roughly 2.5–3 times the staple length	Floating fibres moving in clumps: roller setting too wide for the staple, insufficient fibre control in the main zone, top-roller weighting too light
Roller lapping	Fibre wraps around a top or bottom roller instead of passing forward; delivery breaks down, roller covering is damaged	Setting shorter than the fibre length, damaged or sticky cot surface, excessive humidity/static, broken sliver end carried round, insufficient weighting or clearing
Periodic thick places (spectrogram peaks)	A repeating fault at a fixed sliver length interval, traceable to one rotating part	Eccentric or grooved top roller, bent bottom roller, damaged gear tooth — the faulty part's circumference sets the repeat length

Sliver breaks / ends down	Delivery stops, or a thin missing strand reaches the can	Feed can empty or tangled, web break at the front roller, trumpet choked with fly
Heavy or light sliver (off-weight)	Whole cans off target weight; yarn count drifts	Wrong draft setting, feed sliver off weight, autoleveller fault, wrong number of doublings fed

Fault finding is systematic: weigh delivered sliver regularly (catches off-weight faults), examine evenness traces and spectrograms (a periodic fault's wavelength identifies the faulty roller by its circumference), and check settings, cots and weighting whenever irregularity rises. Later processes only draft this sliver thinner, so a draw-frame fault is magnified, never diluted — the finisher frame is the last point at which sliver evenness can still be improved.

Glossary

Doubling — feeding several slivers together into one delivery so irregularities average out. **Draft** — feed weight (or back surface speed) divided by delivery weight (or front surface speed). **Break draft** — the small draft in the back zone. **Main draft** — the large draft in the front zone. **Floating fibre** — a fibre gripped by neither nip, moving only by friction with neighbours. **Drafting wave** — periodic irregularity from floating fibres moving in clumps. **Roller setting** — nip-to-nip distance between successive pairs, set slightly above staple length. **Cot** — the synthetic-rubber covering of a top roller. **Weighting** — the load pressing top rollers down (dead weight, spring or pneumatic). **Leading / trailing hook** — a fibre hook pointing in / against the direction of travel. **Autoleveller** — a device measuring sliver thickness and varying draft automatically to hold delivered weight constant. **Roller lapping** — fibre wrapping around a roller instead of passing forward.

Key points

• The draw frame drafts, doubles, blends and parallelises: its evenness comes from doubling, not from drafting alone.

• Passage: cans → guide rollers → drafting system → web → trumpet → calendar rollers → coiler and can.

• Total draft = break draft × main draft; the break draft is small (about 1.1–1.5), the main draft does the attenuation.

• Roller setting rule: slightly longer than staple length (staple + about 4–10 mm) — closer breaks fibre, wider loses control of floating fibres.

• Doubling 6 or 8 ends cuts feed variation roughly by √n; draft = (doublings × feed weight) ÷ delivery weight.

• Drafting straightens trailing hooks; each passage reverses hook presentation, so a second passage

straightens the hooks the first one missed.

• The classic defects are sliver irregularity, drafting wave and roller lapping — traced by weight checks, evenness traces and spectrograms.

Practice

1. Six slivers, each of 5.2 g/m, are doubled at a draw frame and delivered as a 5.0 g/m sliver. Calculate the total draft, showing the feed-weight step.
2. One delivery of a draw frame runs at 750 m/min delivering 4.8 g/m sliver at 85 % efficiency. Calculate the actual production in kg/h per delivery, showing each step.
3. A cotton of 32 mm staple length is to be drafted on a 3-over-3 draw frame. Using the setting rule with a 6 mm margin for the main zone, state the main-zone roller setting, and state in one sentence what happens if the setting is made 28 mm instead.
4. Card sliver has a CV of 4.0 %. Estimate the CV of the combined feed when 8 such slivers are doubled, showing the formula and the arithmetic. State one kind of variation doubling cannot correct.
5. Using leading and trailing hooks, explain why a finisher passage straightens hooks the breaker passage left, although both machines draft in the same direction of travel.
6. A finisher sliver shows a regular wave of wavelength near 80 mm on 28 mm staple cotton. Name the defect, calculate the wavelength as a multiple of staple length, and give two setting or weighting causes to check first.

Answer key

1. Total feed = 6 × 5.2 = 31.2 g/m. Draft = 31.2 ÷ 5.0 = **6.24**.
2. Theoretical length per hour = 750 × 60 = 45,000 m. Theoretical mass = 45,000 × 4.8 = 216,000 g = 216 kg/h. Actual = 216 × 0.85 = **183.6 kg/h per delivery**.
3. Main-zone setting ≈ 32 + 6 = **38 mm**. At 28 mm — shorter than the 32 mm staple — fibres are gripped by both nips at once and are broken or dragged, causing breakage, neps and lapping.
4. CV of combined feed ≈ CV ÷ √n = 4.0 ÷ √8 = 4.0 ÷ 2.828 ≈ **1.41 %**. Doubling cannot correct long-term variation (for example a whole can of heavy sliver) nor periodic faults that are in phase in all the feeds.
5. Drafting straightens only hooks that trail at the front nip. Coiling and re-feeding reverse the sliver end-for-end, so hooks that led at the breaker passage trail at the finisher, whose front nip grips the fibre's straight part and pulls the hook out straight. Two passages thus straighten both hook populations.

6. The defect is **drafting wave**: 80 ÷ 28 ≈ **2.86** — about 2.9 times the staple length, inside the classic 2.5–3 range. Check first: (i) roller setting too wide (floating zone too long); (ii) top-roller weighting too light or cots worn, giving poor main-zone control.
